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CMPE 252, Sec 01

Professor Liu

Homework 2

Implementation

Algorithm Implementation:

For this part, I switched from “shortest path from the start” to a cost-to-go approach that thinks backwards from the goal. I treat the graph as directed (only the edges given in the file) and run Dijkstra starting at the goal on the reversed edges. That gives me, for every node, the best remaining cost to reach the goal. With those values in hand, I rebuild the actual path from the start by always picking the next neighbor that gives the smallest total cost (and if there’s a tie, I pick the smallest index to keep it consistent). The program then writes two lines: the list of vertices on the best path, and the list of cost-to-go values for all vertices in order. This matches the assignment’s output format and makes the result easy to check.

Compilation:

My project requires the following command to run:

```
python3 017157582.py
```

After running, it should generate a .txt with the given ID.